User Manual

Raspberry Pi & Smartwatch App

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1. How to Use the App

Smartwatch

1. Charge the smartwatch battery.
2. Turn on the smartwatch by holding the side button.
3. Press the button once to start a hike.
4. View hike statistics directly on the smartwatch during the hike.
5. Press the button again to stop a hike session:
6. Store the hike session:
    a. Move within range of the Raspberry Pi (approx. 2 meters).
    b. Wait until a Bluetooth connection is established.
    c. After "Sending session data to the hub" message has disappeared on the smartwatch, data transfer is successful.
    d. If connection issues occur or "Sending session data to Hub" remains visible, see **Troubleshooting 1**.
    e. Press the button again to start a new hike (this overwrites the previous hike session).
7. Turn off the smartwatch by holding the side button.

Raspberry Pi

1. Start the Raspberry Pi by connecting it to a power source.
2. The application server will start automatically.
    a. If not, see **Troubleshooting 2**.
3. Server establishes a connection to smartwatch via Bluetooth.
    a. If unsuccessful, see **Troubleshooting 1**.
    b. If successful,
        i. The pending hike session is transferred to the server and stored in the database.
        ii. Invalid sessions are discarded, see **Troubleshooting 3**.
4. View hike statistics through the Web UI (see below).
5. Restarting the Raspberry Pi (unplugging and reconnecting power) may help resolve issues.
6. Turn off Raspberry PI by unplugging power.

Web UI

1. Connect your device to the same network as the Raspberry Pi.
2. Find the Raspberry Pi's IP address:
    a. Windows:
        i. Open Command Prompt by typing "cmd" in the search bar and pressing Enter. ii. Type: ipconfig

iii. Locate and copy the IPv4 address. Select IPv4 address and press right mouse button.

b. Linux:
    i. Open Terminal. (press “Ctrl + Alt + T”)
    ii. Type: “ip addr show” iii. Locate and copy the IPv4 address (look for a line starting with “inet”)

3. Open a web browser and enter the IP address with the port:
    a. Example: “12.163.104.178:5000” or simply “IPv4:port”.
4. The Web UI should load.
    a. If not successful, look **Troubleshooting 4**.

2. Useful tips

1. Restarting the Raspberry Pi can resolve:
    a. Bluetooth connection issues.
    b. Server-related problems.
2. Restarting the smartwatch (holding the side button) can fix freezing or unresponsiveness.
3. Bluetooth connection and data transfer is successful, when smartwatch does not show “Sending session to Hub”.

3. Troubleshooting

1. **The smartwatch is stuck connecting via Bluetooth**
    a. Move close to Raspberry Pi.
    b. Wait for the connection to be established.
    c. If it takes long time:
        i. Restart the Raspberry Pi every 15 minutes until successful Bluetooth connection.
        ii. Restart the smartwatch (lose session data).
2. **Web UI server does not start automatically**
    a. Open Terminal (Linux) in folder path and run:
        i. “python3 wserver.py”
        ii. ”python3 receiver.py”
3. **Invalid hike session**
    a. Invalid sessions cannot be recovered.
    b. Report the issue to the application administrators.
4. **Web UI issues**
    a. Verify the IP address:
        i. Ensure you are using the correct IPv4 address.
    b. Verify the correct port:
        i. Check project configuration files for the correct port number.
    c. Restart the Raspberry Pi.

4.Test plan

Sequence of actions testing all functionalities

Smartwatch:

- Power up the smartwatch
    - Smartwatch UI should be visible

- Start a hike session
    - Hike values should be updated and be visible

- Stop a hike session
    - Discard hike session and start a new hike session
    - Stop the new hike session
    - Raspberry Pi is running at this stage
    - Bluetooth connection works with Raspberry Pi. If not,
        - Wait until Bluetooth connection is established.

    - Hike session is sent via Bluetooth. Nice!
- Power down the smartwatch

Raspberry Pi:

- Power up the Raspberry PI
- Open terminal and run:
    - “python3 wserver.py”
    - “python3 receiver.py”
- Bluetooth connection is active
- Get session data via Bluetooth
    - Check session data transfer via web UI
    - If on WebUI,
        - Database has received session data
        - Test that database has received right values
- Power down the Raspberry Pi

WebUI:

- Power up Raspberry Pi and run the above commands
- Connect to same network as Raspberry Pi
- Open a browser
- Get IPv4 address
- Go to WebUI address in browser
    - Port is 5000
- Check site visibility and components
    - Total statistics are visible
    - All single hiking sessions are visible
    - Remove button visible and functional
        - If hike session is not visible after button press, hike session is removed from the database.
- Testing is done

5.Notes

- Starting a new hike overwrites previous data stored on the smartwatch.
- Ensure devices remain within Bluetooth range (approx. 2 m) during synchronization.